

Joint modeling of social genetic effects in mono- and pluri-specific groups: case study in intercrops

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Supplementary File 1

1 Simulations of field trials

1.1 Default parameter values

- Mean yield of wheat genotypes (in q/ha): $\mu^{(f)} = 65$ in sole crops, and reduced by half in intercrops (65×0.5)
- Mean yield of pea genotypes (in qt/ha): $\mu^{(t)} = 30$
- Wheat DBV variance: $\sigma_D^{2(f)} = (CV_g^{(f)} \times \mu^{(f)})^2$ where $CV_g^{(f)} \in \{8\%, 8.5\%\}$
- Prop. of wheat $SBV^{(f,IC)}$ variance: $\sigma_S^{2(f,IC)} = 0.2 \sigma_D^{2(f)}$
- Prop. of wheat SIGV variance: $\sigma_{SIGV}^{2(f)} = 0.5 \sigma_D^{2(f)}$
- Correlation between wheat DBV and $SBV^{(f,IC)}$: $\rho_{DS}^{(f)} = -0.8$
 - Covariance: $\sigma_{DS}^{(f)} = \rho_{DS}^{(f)} / (\sigma_D^{(f)} \times \sigma_S^{(f)})$
- Variance of $(DBV \times SBV)^{IC}$: $\sigma_{D \times S}^{2(f)}$ (resp., $\sigma_{D \times S}^{2(t)} = 0$)
- Variance of wheat BV^{SC} : $\sigma_{BV^{SC}}^{2(f)} = \sigma_D^{2(f)} + \sigma_{SIGV}^{2(f)}$
- Variance of wheat BV^{IC} : $\sigma_{BV^{IC}}^{2(f)} = \sigma_D^{2(f)} + 2 \sigma_{DS}^{(f)} + \sigma_S^{2(f,IC)}$
- Sole error variance for focal: $\sigma_{e,sc}^{2(f)} = \frac{1-h_{sc}^{2(f)}}{h_{sc}^{2(f)}} \times (\sigma_{BV^{SC}}^{2(f)} \times r)$ where $h_{sc}^{2(f)} = 0.7$, and r , the number of replicates (2)
- Intercrop error variance was approxiamte for both, tester and focal by:
 $\sigma_{e,ic}^2 = \frac{1-h_{ic}^2}{h_{ic}^2} \times (\sigma_{BV^{IC}}^2 \times (t \times r))$ where $h_{ic}^2 = 0.7$, t the numer of effective tester, and r the number of replicates
- Correlation between mixed focal and tester errors: $\rho_{e,ic}^{(f,t)} = -0.2$